

Avishake Adhikary

Machine Learning Engineer | Industry Trainer

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📅 15th March 2001

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📁 WORK EXPERIENCE

Machine Learning Engineer

Minion Technologies

June 2025 – Present | Kolkata, India

- Develop and fine-tune medical AI and large language models using **PyTorch**, **LoRA/QLoRA**, and **Unsloth** for healthcare-specific NLP and diagnostic applications.
- Adapt and optimize state-of-the-art open-source LLMs (e.g., **DeepSeek R1**, **Gemma**, **MedGemma-27B**) for domain-specific medical data understanding and inference.
- Quantize large-scale models (**MedGemma-27B-IT**) using **MXFP4** precision optimized for **NVIDIA Blackwell** architecture, improving inference efficiency on modern and edge hardware.
- Design and implement distributed training pipelines using HPC infrastructure and Distributed Data Parallel (**DDP**) across multi-GPU environments.
- Build and optimize data pipelines for large-scale medical datasets (**OpenFDA**, **ICD-11**), including automated extraction, preprocessing, and validation.
- Enhance retrieval-augmented generation (**RAG**) systems by integrating domain-specific knowledge bases, achieving a **30%** improvement in retrieval latency and performance.
- Develop AI-powered content generation pipelines using open-source multimodal models (text, image, video), deployed on cloud infrastructure (Scaleway).

Computer Science Faculty

Bharatiya Vidya Bhavan

February 2025 – May 2025

- Instructing BBA students in foundational computer science subjects, including Graphics Design, SQL, and Image Processing, with a focus on both theoretical concepts and practical applications.
- Leading lab sessions for a class of 39 students, providing hands-on experience and personalized support to enhance their technical skills.
- Grading assignments, projects, and exams, offering constructive feedback to improve student performance and understanding.
- Developing engaging lesson plans and course materials tailored to the needs of diverse learners, fostering an interactive and inclusive classroom environment.

🧩 CORE SKILLS

- **Programming:** Python, Java, JavaScript (React.js, Angular), C++, C, Flutter
- **Machine Learning:** Deep Neural Networks, PyTorch, TensorFlow, Stable Diffusion, Computer Vision
- **Cloud Platforms:** AWS, Azure, Google Cloud
- **Databases & APIs:** MySQL, PostgreSQL, Firebase, SQLite, REST APIs, JSON, XML
- **Web Development:** Node.js (Express.js), HTML, CSS, PHP
- **Other Tools:** Git (CI/CD Pipelines), Bitbucket, Unix, Shell Programming, Computer Architecture, Operating Systems, MS Office

🎓 EDUCATION

Masters in Computer Applications

Amity University Kolkata

September 2021 – June 2023 | Kolkata, India

8.28 CGPA

Roles: Class Representative, Event Manager

Bachelors in Computer Applications

Amity University Kolkata

2018 – 2021 | Kolkata, India

8.21 CGPA

Roles: Class Representative, IT Club Admin

📄 TOP CERTIFICATES

- AWS Computer Vision: Getting Started with GluonCV
- Getting Started with AWS Machine Learning
- Microsoft Certified: Data Analyst Associate with Power BI
- Google Cloud Platform Fundamentals for AWS Professionals
- Industrial IoT on Google Cloud Platform

📁 PROJECTS

AI Job Assistant

November 2023 – present

- Developed a full-stack AI solution utilizing **OpenAI's API** with a custom knowledge base, enhancing data retrieval efficiency by 30%.

Machine Learning Engineer | Graduate Engineer

PTS Consulting Services

August 2023 | Kolkata, India

- **Optimized deep learning workflows** by implementing the **VGG16 model**, improving IDPUI project comparison accuracy by **15%**.
- Designed and deployed a secure **Single Sign-On (SSO)** system with **Okta**, enhancing authentication speed and user experience for the Yujj App.
- Spearheaded UI development for **KoyoX**, an AI job assistant, ensuring seamless integration with **OpenAI APIs** to deliver tailored job recommendations.
- Independently developed the full-stack architecture for an **AI Job Assistant**, integrating a custom knowledge base to increase data retrieval efficiency by **30%**.
- Conducted code reviews and implemented scalable solutions to reduce deployment time by **25%**.

Web Developer Intern

Travarsa Private Limited

July 2022 – August 2022 | Kolkata, India

- Delivered a **full-stack website for MediFitServe**, improving user engagement by implementing responsive design and seamless file upload functionality.
- Conducted **SEO/SEM analysis**, increasing site traffic by **40%** through strategic keyword targeting and optimization.
- Streamlined deployment processes to custom hosting servers, reducing project completion time by **20%**.

Web Designer and Digital Marketing Analyst

Travarsa Private Limited

April 2020 – June 2020 | Kolkata, India

- Engineered a user-centric full-stack website for **Guide Me Wide**, incorporating dynamic pop-ups and file upload capabilities, boosting user retention by **25%**.
- Designed and executed **digital marketing campaigns**, resulting in a **50%** improvement in search engine rankings.
- Automated deployment workflows to custom servers, enhancing site performance and accessibility.

PUBLICATIONS

Machine Learning (ML) and Blockchain for IoE: A Deep Insight into Framework, Security, and its Applications with Industrial IoE

Routledge - Taylor & Francis Group

29th September 2023

Deep Learning for COVID-19

Springer International Publishing

Development on Risk based Information Security Standard for Adaptation of Blockchain enabled systems

Springer International Publishing

Design of an IoT-Based Mobile Healthcare Unit Using a Self-Regulated React.js Database and Arduino

Apple Academic Press

- Engineered both front-end and back-end systems, ensuring scalable and robust performance.

KoyoX - AI Job Assistant UI Development

November 2023 – present

- Led the UI development of **KoyoX**, an AI-powered job assistant, ensuring seamless user experience and functional integration.

Neural-Networks-From-Scratch

January 2024 – present

- Built foundational neural networks, including derivatives, multi-layer perceptrons, and a GPT model with **self-attention** and a decoder-only architecture.
- Designed specialized notebooks like **"NameWeave,"** which transitions from traditional models to neural networks, and **"GPT Tokenizer,"** focusing on text preprocessing and tokenization.

Yujj

October 2023 – present

- Integrated **Okta's Single Sign-On (SSO)** for enhanced security and user authentication.
- Designed and implemented **QR code generation** for seamless data exchange and secure authentication processes.

Text-To-Image Generator

May 2023

- Developed a GUI application for **image generation from text prompts** using the **Stable Diffusion pipeline**.
- Managed dependencies, utilized **PyTorch**, and integrated with **HuggingFace API Token** for seamless functionality.

Face Recognition Flutter

February 2021

- Built a full-stack application with **Dart** and integrated APIs like **MLVision**, **FirebaselImage**, and **FlutterCamera** for robust face recognition.
- Enhanced performance using **GPU Delegates** and employed **PathProvider API** for secure data storage.

Med-It-Yourself

June 2020 – July 2020

- Designed a native Android health app featuring both user and provider interfaces.
- Integrated third-party APIs such as **Google Maps API** for location-based services and **Glide** for efficient image handling.

LANGUAGES

English	
Bengali	
Hindi	
German	
Spanish	